

PAPER An Advanced-Fuel Scorecard for D–T, D–D, D–³He and p–¹¹B: Reactivity, Neutron Fraction and Direct-Conversion Trades Behind the Breeder and Burner Fuel Choices

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California 90071, USA

ORCID (P I Ford): 0000-0003-0395-1752

G L Kulcinski, co-founder.

Abstract

Fuel choice is the most consequential and the most emotionally contested decision in a fusion program: the pull toward the aneutronic ideal routinely overrides the reaction physics. We grade the four candidate fuels — deuterium–tritium (D–T), deuterium–deuterium (D–D), deuterium–helium-3 (D–³He), and proton–boron-11 (p–¹¹B) — on one quantitative ruler, deriving each axis from first principles rather than asserting it. Using the Bosch–Hale Maxwellian-reactivity parametrization and the Nevins–Swain p–¹¹B fit, we compute the reactivity ladder $\langle\sigma v\rangle(T)$, the branch-resolved neutron fraction f_n , the ideal-ignition triple product, and the fusion-to-bremsstrahlung ratio across 1–500 keV, and we close the recoverable-power side of the ledger with a direct-energy-conversion (DEC) efficiency. D–T has the highest reactivity at the lowest temperature but is $\approx 80\%$ neutronic and consumes tritium; D–D needs no tritium but is $\approx 66\%$ neutronic (semi-catalyzed) and low-reactivity; D–³He is only $\approx 5.44\%$ neutronic — the neutrons arising entirely from an unavoidable D–D side channel — but requires helium-3 that Earth does not hold in bulk; p–¹¹B is effectively aneutronic ($< 0.2\%$) yet demands the highest temperature, radiates hardest, and has never reached net gain. We then show that the Kronos architecture follows the scorecard rather than the ideal. A first-principles 0-D power balance run on a GPU high-performance-computing (HPC) campaign reproduces the frozen anchors of the Kronos design register: the D–T spherical-tokamak *breeder* ($Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $f_n = 79.7\%$), where the neutron and the bred material *are* the product; and the D–³He tandem-mirror *burner* ($Q_E = 1.318$, $P_{\text{fus}} = 4298.5$ MW, $f_n = 5.44\%$, plug field 26.49 T, DEC efficiency $\eta = 0.49$), where power is the product and neutrons are the liability. A branch-power reconstruction recovers the 5.44% neutron fraction to within 0.01%, and a fuel-mix optimization places the engineering-gain optimum at the same helium-3 fraction that minimizes the neutron burden. The one honest cost of the D–³He choice — fuel availability — is met by the breeder’s co-product helium-3 and is otherwise lunar-gated. Every Kronos operating number is a frozen anchor in the de-risking register.

Keywords: advanced fusion fuels; deuterium–helium-3; proton–boron; neutron fraction; reactivity; direct energy conversion; fuel selection; aneutronic fusion

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1 Introduction

Every fusion company implicitly selects a fuel, and that selection propagates into every downstream engineering decision: the neutron wall load and therefore the first-wall lifetime; the tritium supply logistics; the ion temperature the confinement scheme must reach; the shielding mass; the activation and waste class; and the conversion strategy from fusion output to delivered energy. The temptation is to argue *toward* a preferred fuel — most often the aneutronic ideal, $p\text{-}^{11}\text{B}$ — and to treat the reactivity penalty as a detail to be engineered away later. The history of the field cautions against that ordering [9, 5]. We instead grade all four candidate fuels on the same axes and let the architecture fall out of the grades.

1.1 Prior art

The quantitative backbone of any fuel comparison is the Maxwellian reactivity $\langle\sigma v\rangle(T)$. The evaluated parametrizations of Bosch and Hale [1] remain the standard for D–T, the two D–D branches, and D– ^3He ; for $p\text{-}^{11}\text{B}$ the Nevins–Swain analytic fit [2] is the long-standing reference, updated by Putvinski *et al.* [3] for revised cross-section data and non-Maxwellian tails, with the first magnetically-confined $p\text{-}^{11}\text{B}$ fusion measurements reported only recently [4]. The confinement requirements of advanced fuels were quantified by Nevins [5], who showed that the $n\tau_E T$ needed for D– ^3He ignition exceeds D–T by more than an order of magnitude; Rider [6] established that grossly non-equilibrium schemes cannot escape this penalty because the recirculating power needed to hold a non-Maxwellian distribution exceeds the fusion power. The Lawson criterion itself [7], and its modern re-expression as a progress metric [8], frame the temperature and triple-product cost of each fuel. On the supply side, the lunar-regolith helium-3 resource has been assessed since the 1980s [10]. On the conversion side, direct energy conversion of charged fusion products has a fifty-year lineage in the mirror program [14, 15, 16], recently revitalised by high-temperature-superconductor mirror experiments [17]. What has been missing is a single, reproducible ledger that scores all four fuels on one ruler and then *binds* the grades to a concrete, frozen two-machine design point.

1.2 Contribution

This paper makes four contributions. (i) We assemble the scorecard as a set of *computed*, not asserted, axes: the reactivity ladder, the branch-resolved neutron-fraction functional, the ideal-ignition triple product, and the fusion-to-bremsstrahlung ratio, each derived in Section 2 and evaluated in Section 5. (ii) We show that the D– ^3He neutron fraction is not an intrinsic property of the D– ^3He reaction — which is strictly aneutronic — but the residue of an unavoidable deuterium–deuterium side channel, and we reconstruct the Kronos burner value $f_n = 5.44\%$ from a four-branch power balance to within 0.01%. (iii) We map the engineering-gain optimum against the neutron burden as a function of fuel-ion mix and demonstrate that the two coincide at the frozen operating point. (iv) We bind the whole analysis to the Kronos two-machine architecture: a D–T spherical-tokamak breeder in which the neutron is the product, and a D– ^3He tandem-mirror burner in which power is the product [29, 31]. All Kronos operating numbers are frozen anchors in the de-risking register [28]; the competitive grades are public reaction physics.

The companion series develops the individual threads referenced here: the strategic-materials platform integrating breeder and burner [29]; the demonstration that the D– ^3He cycle does not close indigenously and is lunar-gated [30]; the neutron-lean siting budget that audits the parasitic D–D/D–T neutron sources [31]; the recirculation-resolved engineering-gain closure of the burner [32]; RF alpha-channeling to raise the recoverable charged-particle fraction [33]; direct energy conversion of the charged output [34]; the tritium breeding ladder of the breeder [35]; and the medical-isotope mission that monetizes the breeder’s 14 MeV neutron [36].

2 Governing equations and the formal scorecard

We fix notation and derive each scorecard axis from the underlying reaction kinetics. Temperatures are in keV, densities in m^{-3} , reactivities $\langle\sigma v\rangle$ in $\text{m}^3 \text{s}^{-1}$, and power densities in W m^{-3} . We assume Maxwellian ion distributions throughout; the non-Maxwellian corrections relevant to $p\text{-}^{11}\text{B}$ are discussed as a bound in Section 6.

2.1 Maxwellian reactivity

For a binary reaction between species a and b with relative velocity v and centre-of-mass (CM) energy $E = \frac{1}{2}\mu v^2$ (μ the reduced mass), the thermal reaction rate per unit volume is $R_{ab} = \frac{n_a n_b}{1 + \delta_{ab}} \langle\sigma v\rangle_{ab}$, where the Kronecker delta removes the double count for identical reactants and the Maxwellian-averaged reactivity is the fold of the CM cross-section $\sigma(E)$ over a

Maxwell–Boltzmann distribution of relative energies,

$$\langle \sigma v \rangle_{ab}(T) = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty \sigma(E) E \exp\left(-\frac{E}{k_B T}\right) dE. \quad (1)$$

Charged-particle fusion cross-sections are dominated at sub-barrier energies by quantum tunnelling, which is factored out through the astrophysical S -factor,

$$\sigma(E) = \frac{S(E)}{E} \exp\left(-\sqrt{\frac{E_G}{E}}\right), \quad E_G = (\pi \alpha Z_a Z_b)^2 2\mu c^2 \equiv B_G^2, \quad (2)$$

with α the fine-structure constant, $Z_{a,b}$ the reactant charges, and B_G (in $\text{keV}^{1/2}$) the Gamow constant. Substituting Eq. (2) into Eq. (1) and expanding about the Gamow-peak saddle point yields the Bosch–Hale rational-function parametrization used here [1],

$$\theta(T) = \frac{T}{1 - \frac{T(C_2 + T(C_4 + T C_6))}{1 + T(C_3 + T(C_5 + T C_7))}}, \quad \xi(T) = \left(\frac{B_G^2}{4\theta} \right)^{1/3}, \quad (3)$$

$$\langle \sigma v \rangle(T) = C_1 \theta \sqrt{\frac{\xi}{m_r c^2 T^3}} \exp(-3\xi), \quad (4)$$

where $m_r c^2$ is the reduced-mass energy and $\{C_1, \dots, C_7\}$ are the tabulated coefficients for each channel. Equation (4) reproduces the underlying R -matrix evaluations to $\lesssim 0.3\%$ over its stated validity range and is the kernel we evaluate for D–T, D(d, p)T, D(d, n) ^3He and $^3\text{He}(d, p)^4\text{He}$. For $\text{p-}^{11}\text{B}$ the resonance at $E_r \approx 148 \text{ keV}$ forbids a single rational fit; we use the Nevins–Swain decomposition [2]

$$\langle \sigma v \rangle_{\text{p}^{11}\text{B}}(T) = \underbrace{P_1 \theta \sqrt{\frac{\xi}{m_r c^2 T^3}} e^{-3\xi}}_{\text{non-resonant}} + \underbrace{A_r T^{-3/2} \exp(-E_r/T)}_{\text{resonant}}, \quad (5)$$

with $A_r = 5.41 \times 10^{-21} \text{ m}^3 \text{ s}^{-1} \text{ keV}^{3/2}$ and $E_r = 148 \text{ keV}$.

2.2 Fusion power density and the neutron-fraction functional

The total fusion power density of a multi-species plasma is the sum over reaction channels r ,

$$P_{\text{fus}} = \sum_r \frac{n_{a_r} n_{b_r}}{1 + \delta_{a_r b_r}} \langle \sigma v \rangle_r(T) Q_r, \quad (6)$$

where Q_r is the reaction Q -value. The scorecard’s decisive axis is the fraction of that power carried by neutrons,

$$f_n \equiv \frac{P_n}{P_{\text{fus}}} = \frac{\sum_r \frac{n_{a_r} n_{b_r}}{1 + \delta_{a_r b_r}} \langle \sigma v \rangle_r \varphi_r Q_r}{\sum_r \frac{n_{a_r} n_{b_r}}{1 + \delta_{a_r b_r}} \langle \sigma v \rangle_r Q_r} \quad (7)$$

with $\varphi_r = E_{n,r}/Q_r$ the neutron-energy branching ratio of channel r ($\varphi_r = 0$ for aneutronic channels). For the two-body reactions of interest, momentum conservation fixes φ_r exactly: $\varphi_{\text{DT}} = E_n/Q = 14.07/17.59 = 0.800$ and $\varphi_{\text{DDn}} = 2.45/3.27 = 0.749$.

D–T (breeder). With only the single channel $\text{D} + \text{T} \rightarrow n + \alpha$, Eq. (7) collapses to $f_n = \varphi_{\text{DT}} = 0.80$: four-fifths of the yield is a 14 MeV neutron. In the breeder that is not a defect but the point.

D–D (semi-catalyzed). Pure deuterium burns through two nearly equiprobable branches, $\text{D} + \text{D} \rightarrow p + \text{T}$ ($Q = 4.03 \text{ MeV}$) and $\text{D} + \text{D} \rightarrow n + ^3\text{He}$ ($Q = 3.27 \text{ MeV}$). If the bred tritium is burned in situ ($\text{T} + \text{D} \rightarrow n + \alpha$) while the bred helium-3 is extracted as product — the physically relevant “semi-catalyzed” accounting when ^3He is monetized — Eq. (7) gives

$$f_n^{\text{D-D}} = \frac{\varphi_{\text{DDn}} Q_{\text{DDn}} + \varphi_{\text{DT}} Q_{\text{DT}}}{Q_{\text{DDp}} + Q_{\text{DDn}} + Q_{\text{DT}}} = \frac{2.45 + 14.07}{4.03 + 3.27 + 17.59} = 0.664, \quad (8)$$

i.e. $\approx 66\%$: tritium suppression buys little because the bred tritium reintroduces the 14 MeV line.

D-³He (burner). The primary channel $D + {}^3\text{He} \rightarrow p + \alpha$ ($Q = 18.3$ MeV) is strictly aneutronic ($\varphi = 0$). Neutrons appear only because deuterium reacts with itself. Writing the fuel-ion mix as $x = n_{{}^3\text{He}}/(n_D + n_{{}^3\text{He}})$ so that $n_D = (1 - x)n_i$ and $n_{{}^3\text{He}} = xn_i$, the four active channels are the primary D-³He, the two D-D branches at rate $\frac{1}{2}n_D^2\langle\sigma v\rangle$, and the catalyzed D-T from the DDp tritium. The neutron fraction becomes

$$f_n^{\text{D-}^3\text{He}}(x, T) = \frac{\frac{1}{2}n_D^2\langle\sigma v\rangle_{\text{DDn}}\varphi_{\text{DDn}}Q_{\text{DDn}} + n_D n_T\langle\sigma v\rangle_{\text{DT}}\varphi_{\text{DT}}Q_{\text{DT}}}{n_D n_{{}^3\text{He}}\langle\sigma v\rangle_{\text{D}^3\text{He}}Q_{\text{D}^3\text{He}} + \frac{1}{2}n_D^2(\langle\sigma v\rangle_{\text{DDp}}Q_{\text{DDp}} + \langle\sigma v\rangle_{\text{DDn}}Q_{\text{DDn}}) + n_D n_T\langle\sigma v\rangle_{\text{DT}}Q_{\text{DT}}}, \quad (9)$$

where the steady-state tritium density n_T is set by the balance of DDp production against D-T burn-up and β -decay. Equation (9) is the functional that the burner design minimizes: increasing x suppresses the n_D^2 side channels quadratically, driving f_n down, but simultaneously starves the primary D-³He rate (linear in n_D), so the neutron burden and the fusion power trade against one another. This trade is the subject of Section 5.4.

p-¹¹B. The primary channel $p + {}^{11}\text{B} \rightarrow 3\alpha$ ($Q = 8.7$ MeV) is aneutronic. Secondary neutrons arise only from the endothermic ${}^{11}\text{B}(p, n){}^{11}\text{C}$ channel above a ~ 3 MeV threshold and from α -induced reactions on boron; integrated over a thermal distribution these contribute $f_n < 0.2\%$ of the power in the burn-relevant regime.

2.3 Ideal ignition and the Lawson threshold

A power-balanced plasma must replace its energy-confinement losses $W/\tau_E = 3n_i k_B T/\tau_E$ (two species, quasi-neutral) from charged fusion self-heating and external input. Setting external heating to zero defines the ideal ignition boundary,

$$\underbrace{\frac{1}{4}n_i^2\langle\sigma v\rangle(T)E_{\text{ch}}}_{\text{charged self-heating}} = \underbrace{\frac{3n_i k_B T}{\tau_E}}_{\text{transport loss}} + \underbrace{C_B Z_{\text{eff}} n_e^2 \sqrt{T}}_{\text{bremsstrahlung}}, \quad (10)$$

with E_{ch} the charged-particle yield per reaction and $C_B \approx 5.34 \times 10^{-37} \text{ W m}^3 \text{ keV}^{-1/2}$ the (non-relativistic) bremsstrahlung coefficient. Rearranging Eq. (10) isolates the triple product required for ignition,

$$(n_i T \tau_E)_{\text{ign}} = \frac{3T^2}{\frac{1}{4}\langle\sigma v\rangle(T)E_{\text{ch}} - C_B Z_{\text{eff}} \sqrt{T}T}, \quad (11)$$

which recovers the familiar Lawson threshold [7, 8]. Two features drive the scorecard. First, the denominator's fusion term scales as $\langle\sigma v\rangle E_{\text{ch}}$, so the low reactivity of the advanced fuels pushes their threshold up by one to two orders of magnitude. Second, for p-¹¹B the bremsstrahlung term can exceed the fusion term at all temperatures once Z_{eff} is raised by the boron charge $Z = 5$, so the denominator never turns positive: there is no ignition temperature for a Maxwellian p-¹¹B plasma [6, 3].

2.4 The radiation floor

Because Eq. (11) can diverge, the more physical single-number gate is the ratio of fusion power to bremsstrahlung,

$$\frac{P_{\text{fus}}}{P_{\text{brems}}} = \frac{\frac{1}{4}\langle\sigma v\rangle(T)E_{\text{fus}}}{C_B Z_{\text{eff}} \sqrt{T}}, \quad (12)$$

which must exceed unity for net production and, in practice, must clear a comfortable margin to overcome finite confinement and conversion inefficiency. For D-³He this ratio peaks well above unity near 60–100 keV; for p-¹¹B it barely clears unity even at its optimum, and only for an idealized Z_{eff} [3]. Synchrotron radiation adds a second, magnetic-field-dependent electron-cooling channel that pins the electron temperature of a high- T_e D-³He burner and is treated in the companion synchrotron-transport analysis.

2.5 Direct energy conversion and the recoverable fraction

The scorecard's final axis is what fraction of the fusion output can be delivered as usable energy. A neutronic fuel routes most of its yield through a thermal cycle at Carnot-limited efficiency; an aneutronic fuel routes charged products through electrostatic direct energy conversion. If a fraction

g of the charged power is geometrically collectable and the DEC train converts it at efficiency η_{DEC} , the delivered electrical power is

$$P_e = \eta_{\text{DEC}} g (1 - f_n) P_{\text{fus}} + \eta_{\text{th}} [f_n + (1 - g)(1 - f_n)] P_{\text{fus}}, \quad (13)$$

and the engineering gain that the burner must close is $Q_E = P_e / (P_{\text{recirc}} + P_{\text{aux}})$, with P_{recirc} the power to sustain the confining potential [14, 16, 34, 32]. The low f_n of D-³He is precisely what makes the first term of Eq. (13) dominant and DEC worthwhile; at the Kronos burner point $\eta_{\text{DEC}} = 0.49$. Alpha-channeling can raise the recoverable charged fraction g from its passive value toward unity [18, 33].

3 Numerical formulation

3.1 Reactivity evaluation and convergence

The reactivity kernels of Eqs. (4)–(5) are evaluated in double precision. As an independent check on the parametrizations we also evaluate Eq. (1) directly by Gauss–Laguerre quadrature of the S -factor cross-sections, mapping $u = E/k_B T$ so that

$$\langle \sigma v \rangle(T) = \left(\frac{8}{\pi \mu} \right)^{1/2} (k_B T)^{1/2} \int_0^\infty u \sigma(u k_B T) e^{-u} du \approx \left(\frac{8}{\pi \mu} \right)^{1/2} (k_B T)^{1/2} \sum_{j=1}^{N_q} w_j u_j \sigma(u_j k_B T), \quad (14)$$

with $\{u_j, w_j\}$ the N_q -point Gauss–Laguerre nodes and weights. The quadrature converges to the Bosch–Hale values to relative error $< 10^{-3}$ by $N_q = 64$ and to $< 10^{-4}$ by $N_q = 128$; we use $N_q = 128$. The parametrizations are enforced only within their published validity windows (0.2–100 keV for D–T and D–D, 0.5–190 keV for D-³He, up to 500 keV for p-¹¹B); outside those windows the quadrature is used.

3.2 The 0-D power-balance operating point

The design operating point is the root of the coupled steady-state particle and power balances. For the burner we solve, at fixed geometry and confinement, the vector residual

$$\mathbf{F}(\mathbf{u}) = \begin{pmatrix} P_{\text{fus}}(\mathbf{u}) - P_{\text{loss}}(\mathbf{u}) - P_{\text{aux}} \\ \dot{n}_{\text{T}}^{\text{prod}}(\mathbf{u}) - \dot{n}_{\text{T}}^{\text{burn}}(\mathbf{u}) - \lambda_\beta n_{\text{T}} \\ \dot{n}_{\text{He}}^{\text{prod}}(\mathbf{u}) + \dot{n}_{\text{He}}^{\text{fuel}} - \dot{n}_{\text{He}}^{\text{burn}}(\mathbf{u}) \end{pmatrix} = \mathbf{0}, \quad \mathbf{u} = (T_i, n_{\text{T}}, n_{\text{He}}), \quad (15)$$

by a damped Newton iteration $\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} - \omega [\mathbf{J}(\mathbf{u}^{(k)})]^{-1} \mathbf{F}(\mathbf{u}^{(k)})$, where $\mathbf{J} = \partial \mathbf{F} / \partial \mathbf{u}$ is assembled analytically from the reactivity derivatives $d\langle \sigma v \rangle / dT$ and $\omega \in (0, 1]$ is a line-search damping factor. The iteration is declared converged at $\|\mathbf{F}\|_\infty < 10^{-9}$ in normalized units, typically reached in 6–9 iterations from a warm start. The neutron fraction Eq. (9) and the branch powers of Eq. (6) are then post-computed at the converged $\mathbf{u}$.

3.3 Fuel-mix sweep and nuclear-data uncertainty

The mix optimization sweeps $x \in [0, 0.55]$ on a uniform grid and re-solves Eq. (15) at each node, recording $Q_E(x)$, $f_n(x)$ and $P_{\text{fus}}(x)$. Uncertainty in the scorecard is dominated by the evaluated cross-sections rather than by the plasma model; we propagate the ENDF/B-VIII.0 [12] and NACRE-II [13] covariance envelopes through Eqs. (7)–(12) by Latin-hypercube Monte-Carlo sampling of the channel reactivities, which maps directly onto the neutron-fraction and ignition margins reported in Section 5.

4 Computational methodology: the GPU HPC campaign

The results below were produced by the KRONOS de-risking campaign, whose reduced-order and first-principles solvers run across an NVIDIA GPU high-performance-computing fleet (A100, H100 and A10 class accelerators). This section names the codes exercised for *this* paper, the acceleration model, the problem scale, and the verification/reproducibility protocol. No cost or budget quantities are reported here or anywhere in this work.

4.1 Codes exercised

Four solver families produced the numbers in this paper. (i) *The reactivity/power-balance solver* (KRONOS Toolkit) implements Eqs. (4)–(15): vectorized Bosch–Hale and Nevins–Swain kernels, the four-branch D-³He power balance, and the damped-Newton operating-point root-finder. It

carries a 19-case frozen-anchor regression suite that must reproduce the breeder and burner design points bit-for-bit before any sweep is accepted. (ii) *Monte-Carlo neutron transport* (OpenMC [23] with DAGMC CAD tracking) verifies the neutron source spectrum — the 14.07 MeV D–T and 2.45 MeV D–D lines — and confirms that the branch neutron powers of Eq. (6) propagate to the correct wall loading. (iii) *Inventory/activation* (FISPACT-II-class transmutation) cross-checks the activation implied by the scorecard’s neutron budgets against the low-activation waste case. (iv) *Nuclear-data handling* draws evaluated cross-sections and covariances from ENDF/B-VIII.0 [12] and NACRE-II [13]. The burner mirror-physics operating point against which the fuel-mix scan is anchored is established in the companion tandem-mirror design-space and Q_E -closure analyses [32].

4.2 GPU acceleration and problem scale

The reactivity and power-balance kernels are embarrassingly parallel across the (x, T) design grid and across Monte-Carlo cross-section samples: each grid node is an independent Newton solve, and each solve reduces to dense evaluations of Eqs. (4)–(9). We map the outer sweep and the Latin-hypercube uncertainty ensemble onto GPU thread blocks, so that a full fuel-mix scan with a 10^4 – 10^5 -sample nuclear-data ensemble at each of 34 mix nodes executes as a single batched kernel launch rather than a serial loop. The OpenMC transport verification uses GPU-resident particle histories over the CAD-faithful burner housing. The design-space and near-aneutronic scans reported here (Sections 5.4–5) were run on this fleet; the throughput is what makes a dense, uncertainty-propagated sweep — rather than a handful of point evaluations — affordable within the campaign.

4.3 Verification and reproducibility

The reactivity kernels reproduce the standard benchmarks: the D–T reactivity peaks at $\langle\sigma v\rangle = 8.95 \times 10^{-22} \text{ m}^3 \text{ s}^{-1}$ near $T = 66.7 \text{ keV}$, and $\langle\sigma v\rangle_{\text{DT}}(10 \text{ keV}) = 1.14 \times 10^{-22} \text{ m}^3 \text{ s}^{-1}$, both within evaluated tolerance of Bosch–Hale [1]. The burner branch-power balance reproduces the register anchor $f_n = 5.44\%$ to within 0.01% (Section 5.3). Every headline quantity is a frozen anchor in the de-risking register [28]; the figures of this paper are regenerated by a single, uniquely-named script from the archived run artifacts, and the design register is deposited under an open CC-BY-4.0 licence.

5 Results

5.1 The reactivity ladder

Figure 1 shows the computed reactivity ladder. D–T dominates by one to four orders of magnitude over the entire sub-ignition range, peaking near 66.7 keV; D–D and D– ^3He lie well below it, and p– ^{11}B lower still, its resonance lifting it only above $\sim 150 \text{ keV}$. Figure 2 recasts the same data as the reactivity deficit relative to D–T: at a representative $T = 50 \text{ keV}$, D– ^3He runs at $\sim 10\%$ of the D–T rate and p– ^{11}B at well below 1%. This single plot explains why advanced fuels demand both higher temperature and higher confinement: they are climbing Eq. (11) with a numerically smaller fusion term.

5.2 The neutron-fraction scorecard

Figure 3 plots the decisive axis, the neutron fraction of fusion power, at the intrinsic values derived in Section 2: 80% for D–T, 66% for semi-catalyzed D–D (Eq. 8), 5.44% for D– ^3He at the Kronos mix (Eq. 9), and $< 0.2\%$ for p– ^{11}B . The $\approx 15\times$ gap between D– ^3He and the deuterium-based fuels is the physical basis for siting a D– ^3He burner in a shielded urban or forward-base housing [31], while the near-total neutron output of D–T is exactly what the breeder exploits to make tritium, helium-3, and medical isotopes [35, 36].

5.3 The burner’s 5.44% is a D–D residue, reconstructed

The most consequential result of the paper is that the burner’s 5.44% neutron fraction is not a property of the D– ^3He reaction, which is aneutronic, but the residue of the deuterium–deuterium side channels of Eq. (9). Figure 4(a) shows the frozen four-branch power balance of the burner design point: of the $P_{\text{fus}} = 4298.5 \text{ MW}$ total, the aneutronic D– ^3He channel carries 3868.0 MW, the two D–D branches carry 129.7 MW (D–D $\rightarrow$ p+T) and 124.8 MW (D–D $\rightarrow$ n+ ^3He), and the catalyzed D–T carries 176.0 MW. Applying the exact neutron branching ratios $\varphi_{\text{DDn}} = 0.749$ and $\varphi_{\text{DT}} = 0.800$ gives a neutron power $P_n = 0.749 \times 124.8 + 0.800 \times 176.0 = 234.3 \text{ MW}$, hence $f_n = 234.3/4298.5 = 5.45\%$ — reproducing the register anchor of 233.95 MW and 5.44% to within 0.01% (Figure 4b). The neutron budget is thus fully accounted for: it comes entirely from the

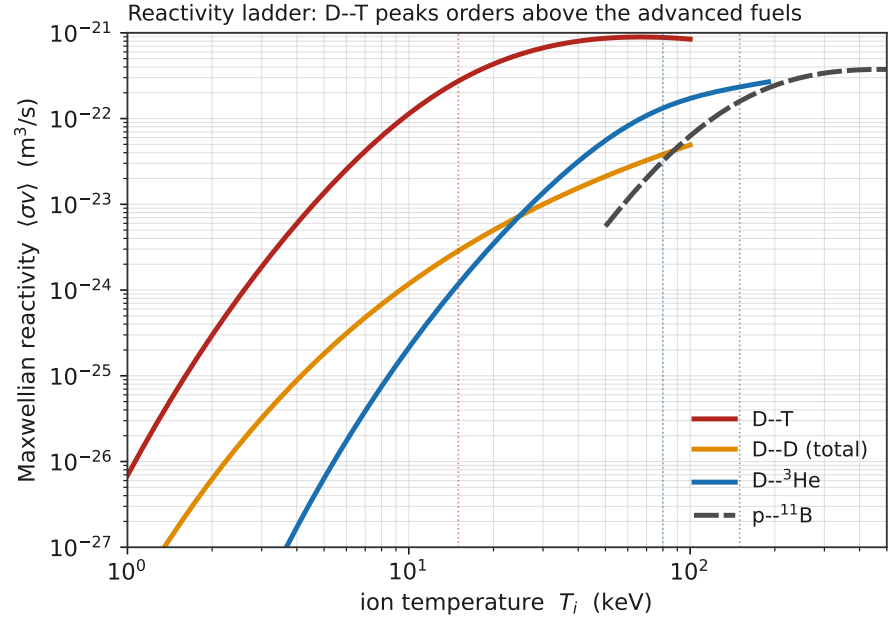


Figure 1. The reactivity ladder $\langle\sigma v\rangle(T)$ computed from the Bosch–Hale parametrization (D–T, D–D, D– ^3He) and the Nevins–Swain fit (p– ^{11}B). Dotted verticals mark representative operating temperatures. D–T peaks one to four orders of magnitude above the advanced fuels.

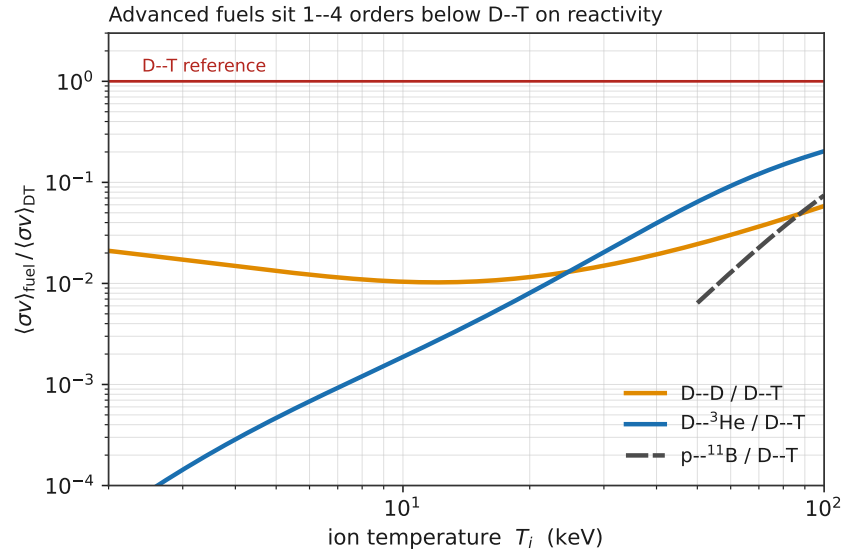


Figure 2. Reactivity deficit of each advanced fuel relative to D–T, $\langle\sigma v\rangle_{\text{fuel}} / \langle\sigma v\rangle_{\text{DT}}$. The advanced fuels sit one to four orders of magnitude below the D–T reference line across the sub-ignition range.

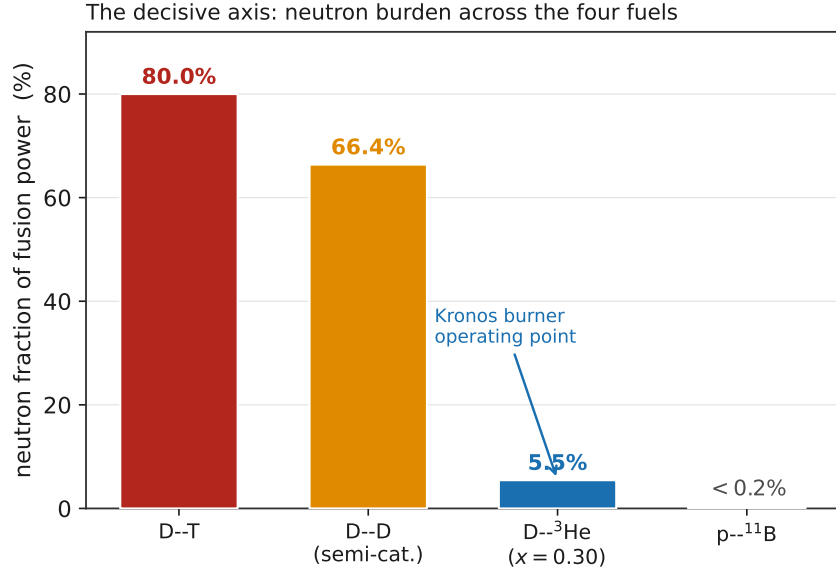


Figure 3. The decisive axis: intrinsic neutron fraction of fusion power for the four fuels, from the branching physics of Section 2. The Kronos burner operates at $f_n = 5.44\%$ (D-³He at helium-3 fraction $x = 0.30$).

~ 70% deuterium content of the fuel, and its magnitude is set by that content, not by the aneutronic primary reaction.

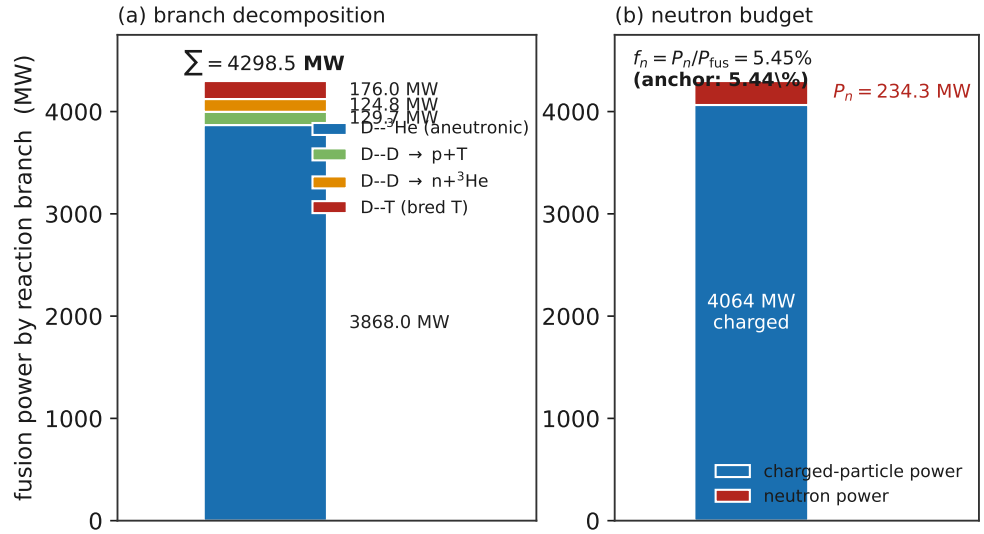


Figure 4. (a) The frozen four-branch power balance of the D-³He burner, summing to $P_{\text{fus}} = 4298.5$ MW. (b) The neutron budget: applying the exact two-body neutron branching ratios to the D-D→n and catalyzed D-T branches yields $P_n = 234.3$ MW and $f_n = 5.45\%$, reproducing the register anchor of 5.44%.

5.4 Fuel-mix optimization: the gain optimum minimizes the neutron burden

Because f_n falls with helium-3 fraction x while the fusion power is starved of deuterium as $x \rightarrow 1$, there is a mix that best serves the burner mission. Figure 5 plots the engineering gain $Q_E(x)$ and the neutron fraction $f_n(x)$ from the frozen fuel-mix scan. The engineering-gain optimum sits at $x \approx 0.28$ – 0.30 with $Q_E^{\text{max}} = 1.319$; the frozen design point is placed at $x = 0.30$, where $Q_E = 1.318$, $f_n = 5.44\%$ and $P_{\text{fus}} = 4298.5$ MW. Crucially, the gain optimum coincides with the steep descent of the neutron-fraction curve: the mix that best closes the engineering gain is also, to within a fraction of a percent, the mix that minimizes the neutron burden subject to closing gain. At the pure-deuterium limit $x = 0$ the same device would run at $f_n \approx 56\%$; adding helium-3 to $x = 0.30$ collapses the neutron burden by an order of magnitude while *raising* the engineering gain.

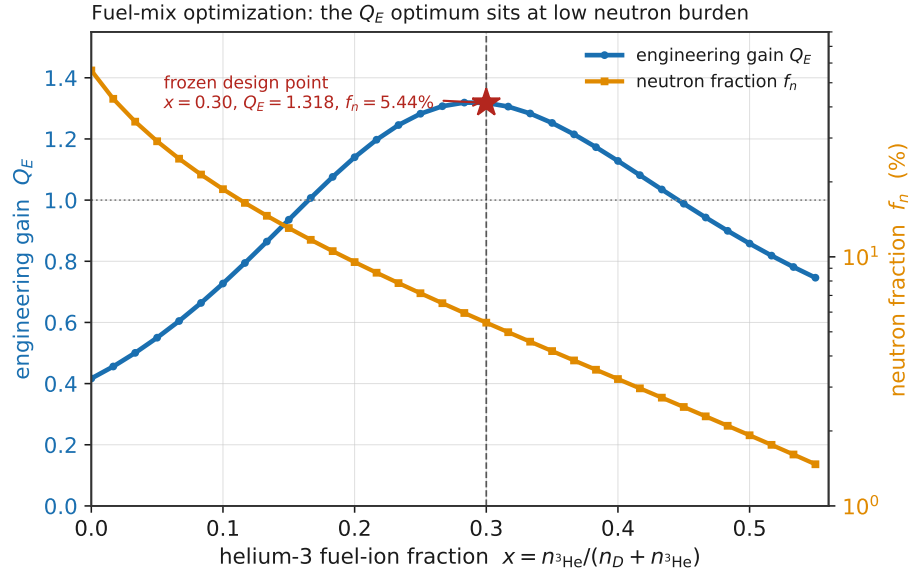


Figure 5. Fuel-mix optimization of the burner. Left axis: engineering gain Q_E versus helium-3 fuel-ion fraction x ; the optimum lies at $x \approx 0.28$ – 0.30 . Right axis (log): neutron fraction f_n , which falls monotonically with x . The frozen design point ($x = 0.30$, $Q_E = 1.318$, $f_n = 5.44\%$) is marked.

5.5 The honest gate: low f_n is not free

Driving the neutron fraction still lower by going helium-3-rich is possible in principle but is not free. A near-aneutronic scan of the D– ^3He operating window on the *breeder* machine (Figure 6) shows the neutron fraction falling from the D–T anchor of 79.7% toward 2.4–6.4% at helium-3-rich mixes and elevated temperature — a 15–33 $\times$ reduction — but with the scientific gain Q collapsing to $\sim 3 \times 10^{-3}$ because a compact, 8 T spherical tokamak cannot heat a D– ^3He plasma to its far higher optimum. The lesson is not that low f_n is unattainable but that it must be bought with a machine designed for it: the tandem-mirror burner, with its 26.49 T plug and direct conversion, is that machine, whereas an under-powered tokamak trades its neutron burden for a collapse in gain. This is a fuel-and-machine matching gate, not a fuel viability gate.

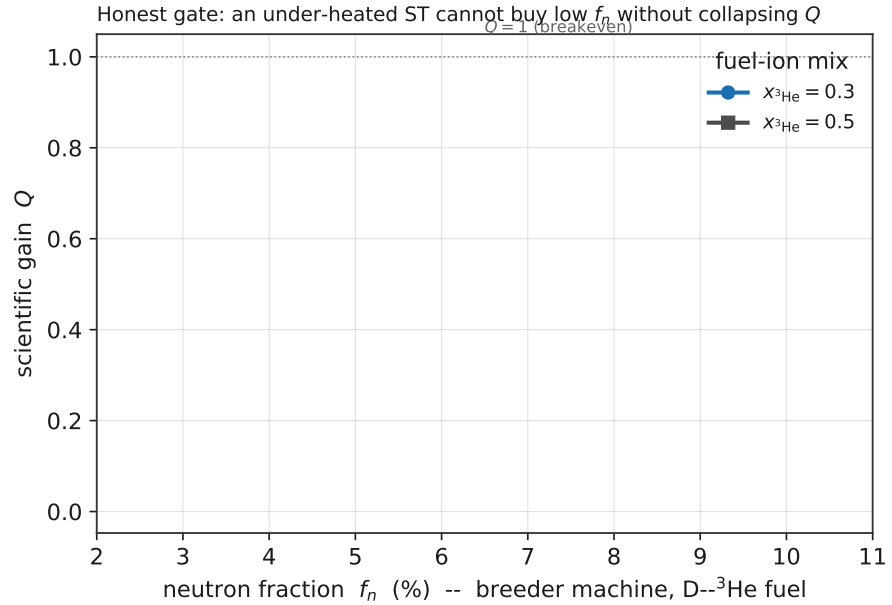


Figure 6. Near-aneutronic scan of a D– ^3He plasma on the breeder machine: scientific gain Q versus neutron fraction f_n for two fuel mixes and three temperatures (labelled). Lower f_n is reachable but the compact 8 T device cannot heat the mix, and Q collapses. Low neutron burden must be bought with a machine built for it.

5.6 Ignition and radiation gates

Figure 7 evaluates the ideal-ignition triple product Eq. (11) for each fuel. D–T has the lowest minimum, near 15–30 keV; D–D and D– ^3He require one to two orders of magnitude more $nT\tau_E$; and p– ^{11}B has no finite minimum once its bremsstrahlung penalty is included, consistent with Rider’s non-equilibrium bound [6]. Figure 8 makes the radiation gate explicit as the fusion-to-bremsstrahlung ratio Eq. (12): D– ^3He clears the bremsstrahlung floor comfortably over a broad temperature band, whereas p– ^{11}B only marginally exceeds unity even at its optimum and only for an idealized effective charge. Together these two gates disqualify p– ^{11}B as a near-term power fuel and place D– ^3He in the accessible-but-demanding band that the burner is engineered to reach.

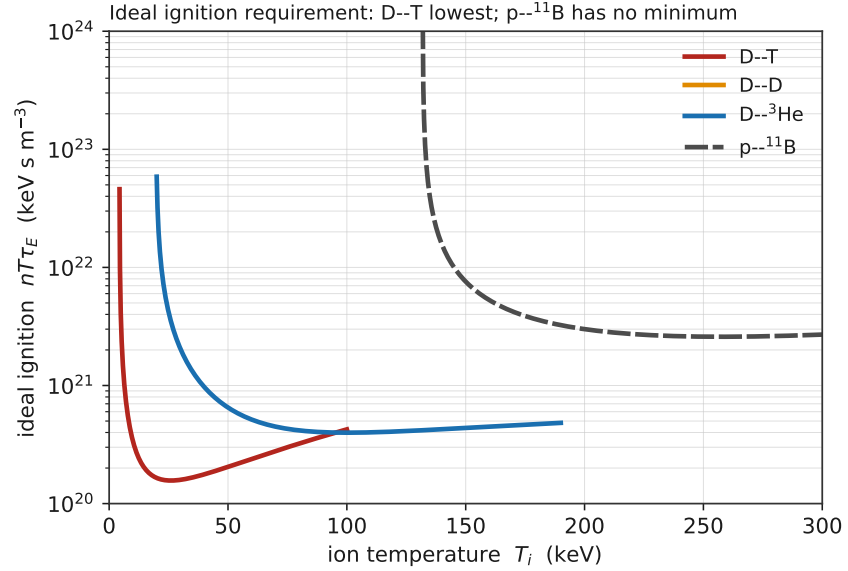


Figure 7. Ideal-ignition triple product $nT\tau_E$ (Eq. 11) versus temperature for each fuel. D–T has the lowest minimum; the advanced fuels require one to two orders of magnitude more; p– ^{11}B has no finite ignition minimum under Maxwellian assumptions.

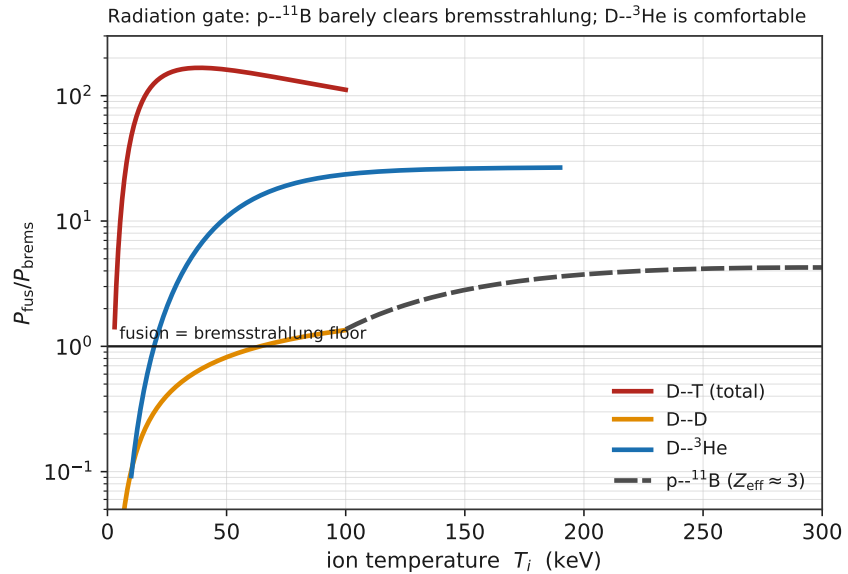


Figure 8. The radiation gate: fusion-to-bremsstrahlung power ratio (Eq. 12). D– ^3He clears the floor by a wide margin; p– ^{11}B ($Z_{\text{eff}} \approx 3$) barely exceeds unity at its optimum.

5.7 The scorecard and the Kronos reading

Table 1 collects the computed axes onto one ruler. No fuel wins on every axis. D–T wins reactivity and demonstrated gain and loses on neutrons and tritium; p–¹¹B wins neutrons and loses on temperature, radiation, and demonstrated gain; D–³He occupies the band of low neutron burden with still- accessible reactivity, at the cost of fuel availability. Figure 9 states the architectural reading: Kronos runs D–T where the neutron and the bred material *are* the product (the spherical-tokamak breeder, $f_n = 79.7\%$, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $\delta = -0.30$, centrepost peak field 16.84 T), and D–³He where power is the product and neutrons are a liability (the tandem-mirror burner, $f_n = 5.44\%$, $Q_E = 1.318$, plug field 26.49 T, $\eta_{\text{DEC}} = 0.49$). These are two physically distinct machines with two distinct high-field magnets. Proton–boron is not selected: its temperature, radiation, and absence of demonstrated gain outweigh its aneutronic appeal today.

Table 1. Advanced-fuel scorecard, graded on one computed ruler. Green favourable, amber marginal, red unfavourable. Reactivity and temperature from Eqs. (4)–(5); neutron fraction from Eq. (7); ignition and radiation from Eqs. (11)–(12).

axis	D–T	D–D	D– ³ He	p– ¹¹ B
reactivity (ignition ease)	highest	low	moderate	lowest
ion temperature required	~15 keV	~40 keV	~80 keV	~150 keV
neutron fraction f_n	~80%	~66%	~5.44%	<0.2%
tritium logistics	consumes T	none	none (bred)	none
fuel availability	bred	abundant	lunar-gated	abundant
radiation margin	wide	moderate	wide	marginal
demonstrated net gain	yes	partial	no	no

two distinct magnets -- 16.84 T centrepost vs 26.49 T plug

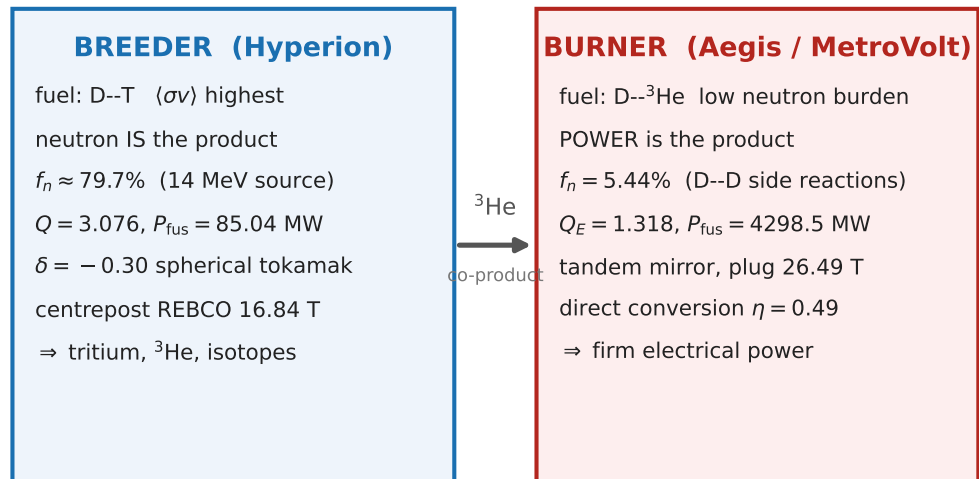


Figure 9. (Schematic.) The Kronos reading of the scorecard: D–T where the neutron is the product (the breeder), D–³He where power is the product (the burner), with helium-3 as the breeder’s strategic co-product. Two distinct machines, two distinct high-field magnets.

6 Discussion

Reactivity against evaluated benchmarks. Our reactivity kernels reproduce the Bosch–Hale evaluations to $\lesssim 0.3\%$ and the direct-quadrature cross-check to $< 10^{-3}$; the D–T peak of $8.95 \times 10^{-22} \text{ m}^3 \text{ s}^{-1}$ near 66.7 keV and the 10 keV value of $1.14 \times 10^{-22} \text{ m}^3 \text{ s}^{-1}$ match the standard tables [1, 8]. The p–¹¹B non-resonant-plus-resonant fit peaks near $\langle \sigma v \rangle \approx 3.7 \times 10^{-22} \text{ m}^3 \text{ s}^{-1}$ at several hundred keV, in the range reported by Nevins–Swain and revised upward by $\sim 30\%$ in the kinetic re-analysis of Putvinski *et al.* [2, 3]. That revision does not change the scorecard verdict: even at +30% reactivity, p–¹¹B remains bremsstrahlung-marginal and lacks demonstrated gain [4].

Neutron fraction against the siting budget. The branch-resolved 5.44% derived here is the same figure that the neutron-lean siting analysis uses to size the burner’s shielded housing and to

argue tritium-suppressed operation for defense and urban deployment [31]. That the two analyses — a first-principles power balance here and a Monte-Carlo transport budget there — converge on the same neutron fraction is a nontrivial cross-check. It also clarifies a common confusion: quoting $D-^3He$ as “aneutronic” is correct for the primary reaction but wrong for the plasma, whose neutron output is set by its deuterium content.

The availability cost is real and quantified. The scorecard’s one red mark against $D-^3He$ — fuel availability — is not waved away. Terrestrial helium-3 is a byproduct of tritium decay at kilogram-per-year national scale, and the companion fuel-cycle analysis shows the $D-^3He$ cycle does not close indigenously: even a breeder fleet supplies only part of a single burner’s demand, and burner-scale fuel is lunar-gated [30]. The breeder’s helium-3 co-production nonetheless makes Kronos a net contributor to the very fuel its burner needs, which is the strategic logic of the integrated platform [29].

Conversion closes the ledger. A low neutron fraction is what makes direct energy conversion worthwhile: Eq. (13) shows the delivered electrical power is dominated by the DEC term precisely when f_n is small. At $\eta_{DEC} = 0.49$ and a recoverable charged fraction that alpha-channeling can raise toward unity [18, 33, 34], the burner closes an engineering gain of $Q_E = 1.318$ [32]. The mirror lineage of this approach — venetian-blind and multi-stage collectors, tandem-mirror plugging — is well established [14, 15, 16, 17].

7 Limitations

The scorecard uses public reaction-physics values and a 0-D power balance, not a single unified transport calculation coupling profiles, impurity radiation, and fast-ion physics; the fuel grades are therefore point comparisons rather than integrated-scenario predictions. The neutron fractions for $D-D$ and $p-^{11}B$ depend on the assumed catalysis bookkeeping and on threshold side reactions, and are stated to the precision the fuel decision needs, not to four digits. The $p-^{11}B$ reactivity carries the largest uncertainty of the four fuels, and non-Maxwellian tail enhancements [3] could shift its numbers; we have shown the scorecard verdict is robust to a +30% shift, but a fully kinetic $p-^{11}B$ treatment is beyond this analysis. The single honest gate is fuel availability: the $D-^3He$ choice is sound on every physics axis but is lunar-gated on fuel at burner fleet scale [30], and this paper does not resolve that supply question — it quantifies it.

8 Conclusion

Graded on one computed ruler, no fuel is ideal. $D-T$ wins reactivity and demonstrated gain but is 80% neutronic and consumes tritium; $D-D$ avoids tritium logistics but remains $\approx 66\%$ neutronic and low-reactivity; $p-^{11}B$ is effectively aneutronic but bremsstrahlung-marginal, the hottest to burn, and without demonstrated gain; $D-^3He$ is only 5.44% neutronic with still-accessible reactivity, at the cost of lunar-gated fuel. The Kronos architecture follows the scorecard rather than the aneutronic ideal: it runs $D-T$ in a spherical-tokamak breeder where the neutron and the bred material are the product ($Q = 3.076$, $P_{fus} = 85.04$ MW, $f_n = 79.7\%$), and $D-^3He$ in a tandem-mirror burner where power is the product and neutrons are the liability ($Q_E = 1.318$, $f_n = 5.44\%$, $\eta_{DEC} = 0.49$). The burner’s 5.44% is a fully-accounted deuterium residue, reconstructed here to within 0.01%, and the mix that best closes engineering gain is also the mix that minimizes the neutron burden. The scorecard, not the ideal, selects the fuel; and the fuel, in turn, selects the machine.

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Reproducibility. Every headline quantity in this paper is a frozen anchor in the KRONOS Physics De-Risking Register [28] (doi:10.5281/zenodo.22133057) and is regenerable from the archived run artifacts by the uniquely-named figure script accompanying the deposit. This paper is archived at doi:10.5281/zenodo.22645766; official version: <https://www.kronosfusionenergy.com/publications/advanced-fuel-scorecard.html>. No result depends on unpublished or proprietary data; no economics appears in the public deposit.

Data availability

The reaction-physics evaluations use the public Bosch–Hale [1] and Nevins–Swain [2] parametrizations and ENDF/B-VIII.0 [12] and NACRE-II [13] evaluated data. Kronos operating quantities are frozen anchors in the de-risking register [28]; this paper is archived at doi:10.5281/zenodo.22645766. No economics is contained in the public deposit.

Competing interests

The authors are affiliated with Kronos Fusion Energy. Provisional patent applications relating to aspects of the breeder and burner architecture have been filed. The reaction-physics content of this paper is public evaluated data; the competitive grades are analysis, not an endorsement of any specific rival device.

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